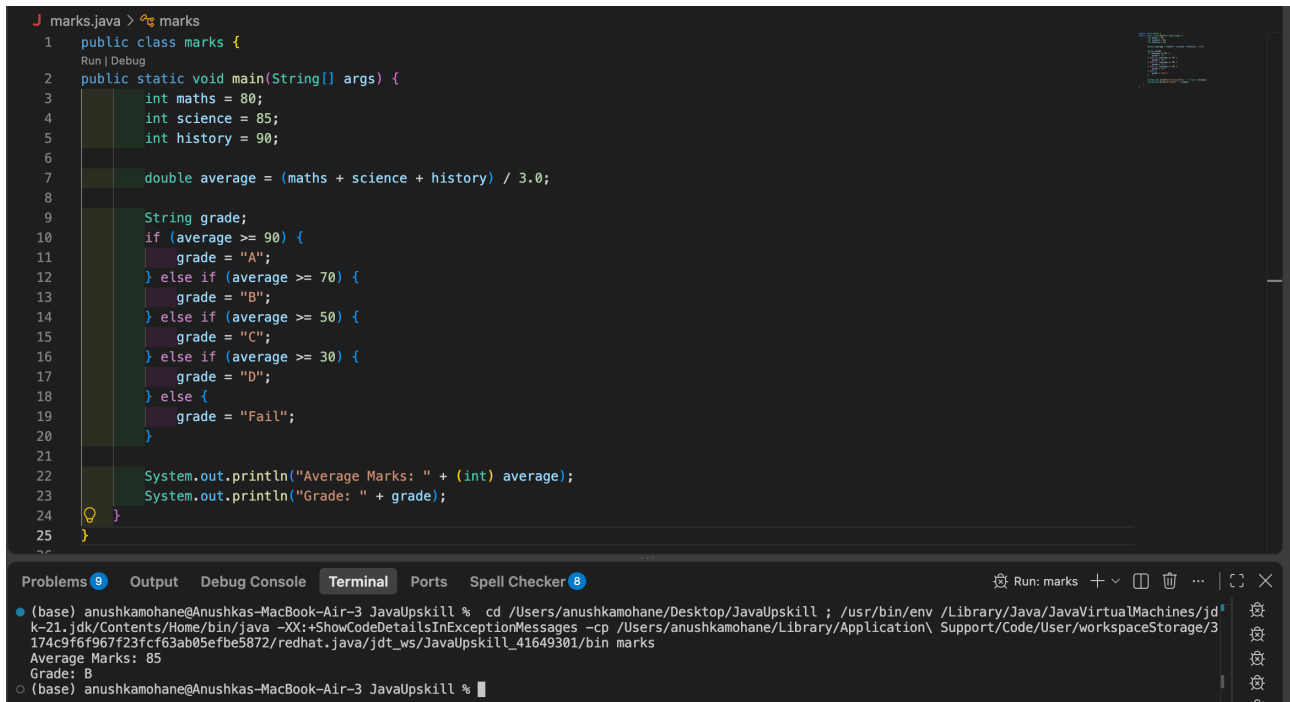


Q1:

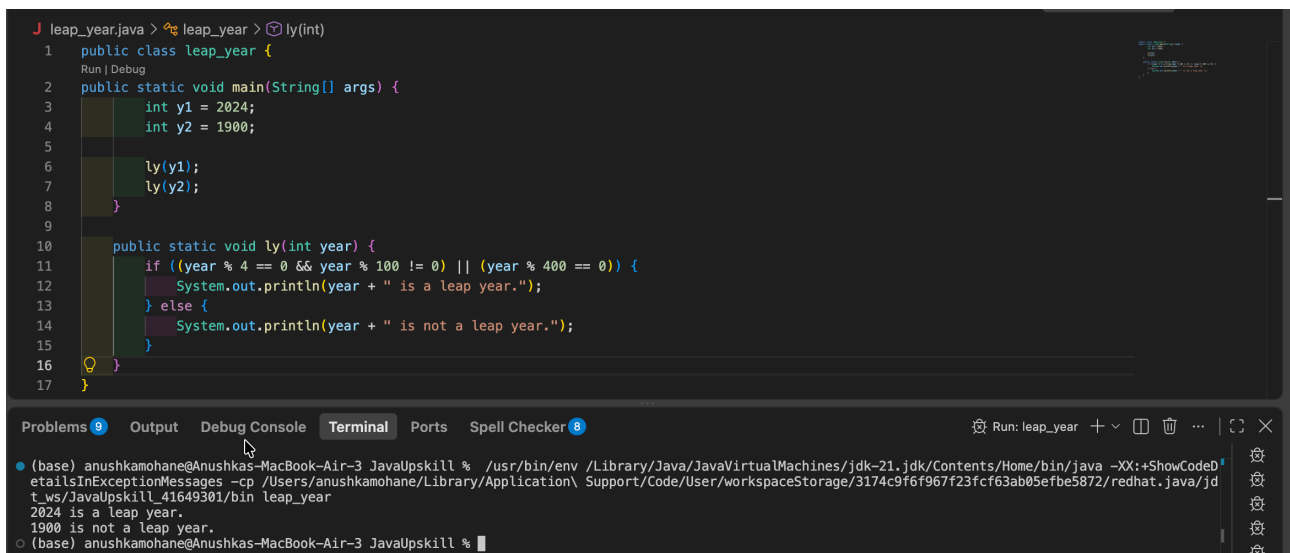
The screenshot shows an IDE with a Java file named `marks.java`. The code calculates the average of three subjects (maths, science, history) and assigns a grade based on the average. The output shows the average marks as 85 and the grade as B.

```
1 public class marks {  
2     public static void main(String[] args) {  
3         int maths = 80;  
4         int science = 85;  
5         int history = 90;  
6  
7         double average = (maths + science + history) / 3.0;  
8  
9         String grade;  
10        if (average >= 90) {  
11            grade = "A";  
12        } else if (average >= 70) {  
13            grade = "B";  
14        } else if (average >= 50) {  
15            grade = "C";  
16        } else if (average >= 30) {  
17            grade = "D";  
18        } else {  
19            grade = "Fail";  
20        }  
21  
22        System.out.println("Average Marks: " + (int) average);  
23        System.out.println("Grade: " + grade);  
24    }  
25 }
```

Terminal Output:

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill % cd /Users/anushkamohane/Desktop/JavaUpskill ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin marks  
Average Marks: 85  
Grade: B  
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

Img 1: Code + Output

Q2:

The screenshot shows an IDE with a Java file named `leap_year.java`. The code checks if a given year is a leap year. The output shows that 2024 is a leap year and 1900 is not a leap year.

```
1 public class leap_year {  
2     public static void main(String[] args) {  
3         int y1 = 2024;  
4         int y2 = 1900;  
5  
6         ly(y1);  
7         ly(y2);  
8     }  
9  
10    public static void ly(int year) {  
11        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {  
12            System.out.println(year + " is a leap year.");  
13        } else {  
14            System.out.println(year + " is not a leap year.");  
15        }  
16    }  
17 }
```

Terminal Output:

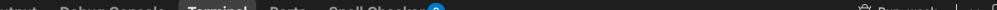
```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin leap_year  
2024 is a leap year.  
1900 is not a leap year.  
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

Img 2: Code + Output

```

1  week.java > week > main(String[])
2  public class week {
3      int day_num = 3;
4      String day_name;
5
6      switch (day_num) {
7          case 1:
8              day_name = "Monday";
9              break;
10         case 2:
11             day_name = "Tuesday";
12             break;
13         case 3:
14             day_name = "Wednesday";
15             break;
16         case 4:
17             day_name = "Thursday";
18             break;
19         case 5:
20             day_name = "Friday";
21             break;
22         case 6:
23             day_name = "Saturday";
24             break;
25         case 7:
26             day_name = "Sunday";
27             break;
28         default:
29             day_name = "Invalid day number";
30             break;
31     }
32
33     if (day_name.equals(anObject: "Invalid day number")) {
34         System.out.println(day_name);
35     } else {
36         System.out.println("The day is " + day_name + ".");
37     }
38 }
39
40
41

```



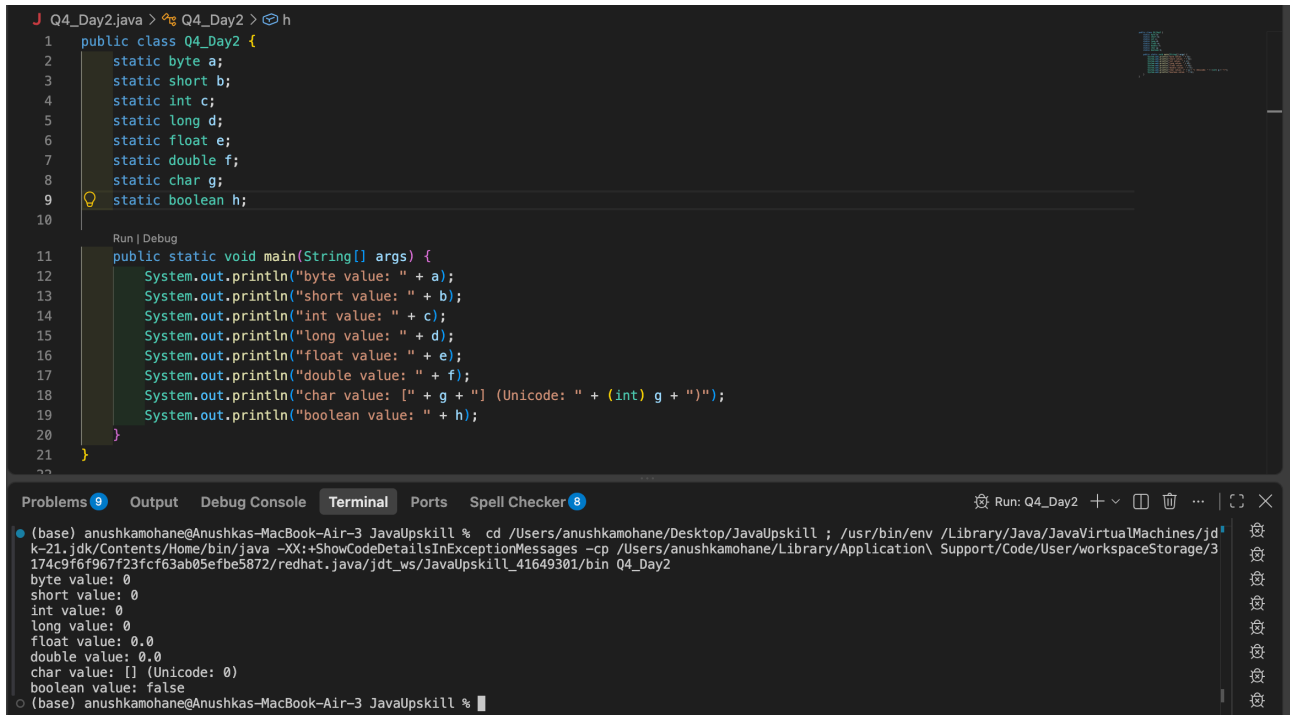
The screenshot shows a terminal window with the following content:

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeD
etailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jd
t_ws/JavaUpskill_41649301/bin week
The day is Wednesday.
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

Img 4: Output

Q4:

Ive printed off the default values.



The screenshot shows an IDE with a Java file named `Q4_Day2.java`. The code defines a class `Q4_Day2` with static variables of various types and a `main` method that prints their default values. The output in the terminal window is as follows:

```
byte value: 0
short value: 0
int value: 0
long value: 0
float value: 0.0
double value: 0.0
char value: [] (Unicode: 0)
boolean value: false
```

The terminal window also shows the command used to run the program:

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill % cd /Users/anushkamohane/Desktop/JavaUpskill ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fc63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin Q4_Day2
```

Img 5: Code + Output